

1. 2번	2. 3번	3. 2번	4. 2번	5. 1번
6. 1번	7. 1번	8. 5번	9. 3번	10. 2번
11. 4번	12. 4번	13. 1번	14. 4번	15. 4번
16. 2번	17. 3번	18. 1번	19. 4번	20. 4번
21. 4번	22. 1번	23. 4번	24. 4번	

1. $\int_0^1 \ln x dx = [x \ln x - x]_0^1 = -1$

2. $\int_{-\frac{2}{3}}^{\infty} \frac{1}{(3x-1)^2+9} dx = \frac{1}{3} \times \frac{1}{3} \left[\tan^{-1} \frac{3x-1}{3} \right]_{-\frac{2}{3}}^{\infty} = \frac{\pi}{12}$

3. $\int_0^{\infty} e^{-x} \cos x dx = \left[\frac{e^{-x}}{2} (-\cos x + \sin x) \right]_0^{\infty} = \frac{1}{2}$

4. $\int_0^1 x \ln x dx = \left[\frac{1}{2} x^2 \ln x \right]_0^1 - \int_0^1 \frac{1}{2} x dx = 0 - \left[\frac{1}{4} x^2 \right]_0^1 = -\frac{1}{4}$

5. $\int_0^{\infty} \frac{1}{(x^2+1)^2} dx = \int_0^{\frac{\pi}{2}} \frac{\sec^2 \theta}{\sec^4 \theta} d\theta \quad (\because x = \tan \theta)$
 $= \int_0^{\frac{\pi}{2}} \cos^2 \theta d\theta = \frac{1}{2} \times \frac{\pi}{2}$

6. $\int_{-\infty}^{\infty} \frac{1}{\pi} \frac{1}{1+x^2} dx = \frac{1}{\pi} [\tan^{-1} x]_{-\infty}^{\infty} = \frac{1}{\pi} \times \pi$

7. $\int_0^{\infty} \frac{1}{e^x+1} dx = \int_1^{\infty} \frac{1}{(t+1)t} dt \quad (\because e^x = t)$
 $= \int_1^{\infty} \frac{1}{t} - \frac{1}{t+1} dt$
 $= [\ln t - \ln(t+1)]_1^{\infty} = \left[\ln \frac{t}{t+1} \right]_1^{\infty} = -\ln \frac{1}{2} = \ln 2$

8. $\int_0^{\infty} \frac{20x}{(x^2+1)(3x+1)} dx = \int_0^{\infty} \frac{2x+6}{x^2+1} + \frac{-6}{3x+1} dx$
 $= \int_0^{\infty} \frac{2x}{x^2+1} + \frac{6}{x^2+1} + \frac{-6}{3x+1} dx$
 $= [\ln(x^2+1) + 6 \tan^{-1} x - 2 \ln(3x+1)]_0^{\infty}$
 $= \left[\ln \frac{x^2+1}{(3x+1)^2} + 6 \tan^{-1} x \right]_0^{\infty} = \ln \frac{1}{9} + 6 \times \frac{\pi}{2}$

9. $\int_{e^3}^{\infty} \frac{dx}{x \ln x (\ln \ln x)^3} = \int_{\ln 3}^{\infty} \frac{1}{t^3} dt \quad (\because \ln \ln x = t, \frac{1}{x \ln x} dx = dt)$
 $= \left[-\frac{1}{2t^2} \right]_{\ln 3}^{\infty} = \frac{1}{2(\ln 3)^2}$

10. $\int_0^{\infty} t^5 e^{-t^3} dt = \frac{1}{3} \int_0^{\infty} x e^{-x} dx \quad (\because t^3 = x) = \frac{1}{3} \times \Gamma(2) = \frac{1}{3} \times 1!$

11. $I = \int_{-1}^2 \frac{1}{x^2} dx$ 는 $p = 2 > 1$ 이므로 발산한다.

12. ㄱ. $\int_0^{\infty} e^{-x} dx = \int_0^{\infty} \frac{1}{e^x} dx$: 수렴

ㄴ. $\int_0^{\infty} \frac{1}{1+e^x} dx \approx \int_0^{\infty} \frac{1}{e^x} dx$: 수렴

ㄷ. $\int_{-\infty}^0 e^{-x} dx \quad (\because -x = t) = \int_0^{\infty} e^t dt$: 발산

ㄹ. $\int_0^e \frac{1}{x} dx$: 발산 ($p = 1$)

13. ① $\int_2^{\infty} \frac{dx}{(x-1)^{3/2}} \approx \int_2^{\infty} \frac{1}{x^{3/2}} dx$: 수렴 ($p > 1$)

② $\int_0^1 \frac{1}{x} dx$: 발산 ($p = 1$)

③ $\int_1^{\infty} \frac{x}{x^2+1} dx \approx \int_1^{\infty} \frac{1}{x} dx$: 발산 ($p = 1$)

④ $\int_0^{\infty} \frac{dx}{\sqrt{x+1}} \approx \int_1^{\infty} \frac{1}{\sqrt{x}} dx$: 발산 ($p < 1$)

⑤ $\int_1^3 \frac{1}{x-2} dx$: 발산 ($p = 1$)

14. ① $\int_1^{\infty} \frac{1}{x} dx$: 발산 ($p = 1$)

② $\int_0^2 \frac{1}{(x-1)^2} dx$: 발산 ($p > 1$)

③ $\int_0^{\frac{\pi}{2}} \sec x dx = [\ln(\sec x + \tan x)]_0^{\frac{\pi}{2}} = \infty$: 발산

④ $\int_{-\infty}^0 x e^x dx \quad (\because -x = t)$
 $= \int_{\infty}^0 t e^{-t} dt = - \int_0^{\infty} \frac{t}{e^t} dt \approx - \int_0^{\infty} \frac{1}{e^t} dt$: 수렴

15. ① $\int_0^{\infty} \frac{1}{1+x^2} dx$ 이 수렴
 $\Rightarrow \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx = 2 \int_0^{\infty} \frac{1}{1+x^2} dx$: 수렴 ($p > 1$)

② $\int_2^5 \frac{1}{\sqrt{x-2}} dx$: 수렴 ($p < 1$)

③ $\int_1^{\infty} e^{-x^2} dx = \int_1^{\infty} \frac{1}{e^{x^2}} dx$: 수렴

④ $\int_1^{\infty} \frac{1+e^{-x}}{x} dx \approx \int_1^{\infty} \frac{1}{x} dx$: 발산 ($p = 1$)

16. ㄱ. $\int_1^{\infty} \frac{\cos^4 x}{x^3+1} dx \approx \int_1^{\infty} \frac{1}{x^3} dx$: 수렴 ($p > 1$)

ㄴ. $\int_0^1 \frac{e^{-x}}{x^2} dx \approx \int_0^1 \frac{1}{x^2} dx$: 발산 ($p > 1$)

ㄷ. $\int_2^{\infty} \frac{x^2}{\sqrt{x^5-1}} dx \approx \int_2^{\infty} \frac{1}{\sqrt{x}} dx$: 발산 ($p < 1$)

17. ① $\int_1^{\infty} \frac{1}{x+\sqrt{x}} dx \approx \int_1^{\infty} \frac{1}{x} dx$: 발산 ($p = 1$)

② $\int_0^1 \frac{\ln x}{x} dx$: 발산 ($p = 1$)

③ $\int_1^{\infty} \frac{1}{x+e^x} dx \approx \int_1^{\infty} \frac{1}{e^x} dx$: 수렴

④ $\int_0^1 \frac{1}{x^2} dx$: 발산 ($p > 1$)

$$18. 가. \int_0^{\infty} \frac{1}{\sqrt{x}} dx = \int_0^1 \frac{1}{\sqrt{x}} dx + \int_1^{\infty} \frac{1}{\sqrt{x}} dx : \text{수렴} + \text{발산} = \text{발산}$$

$$나. \int_0^2 \frac{1}{x-1} dx : \text{발산} (p=1)$$

$$다. \int_0^{\frac{\pi}{2}} \sec x dx : \text{발산}$$

$$라. \int_2^4 \frac{1}{\sqrt[3]{x-2}} dx = \int_2^4 \frac{1}{(x-2)^{\frac{1}{3}}} dx : \text{수렴} (p < 1)$$

$$19. ① \int_0^{\infty} \frac{1}{1+x^2} dx \approx \int_1^{\infty} \frac{1}{x^2} dx : \text{수렴} (p > 1)$$

$$② \int_1^{\infty} e^{-x} dx = \int_1^{\infty} \frac{1}{e^x} dx : \text{수렴}$$

$$③ \int_0^1 \ln x dx = -1 : \text{수렴}$$

$$④ \int_0^1 \frac{\sqrt{x}}{1-x} dx \approx \int_0^1 \frac{1}{1-x} dx : \text{발산} (p=1)$$

$$20. ① \int_1^{\infty} \frac{1}{x+x^2} dx \approx \int_1^{\infty} \frac{1}{x^2} dx : \text{수렴} (p > 1)$$

$$\begin{aligned} ② \int_0^{\infty} \frac{1}{\sqrt{x}} e^{-3x} dx \\ &= \int_0^1 \frac{1}{\sqrt{x}} e^{-3x} dx + \int_1^{\infty} \frac{1}{\sqrt{x} e^{3x}} dx \\ &\approx \int_0^1 \frac{1}{\sqrt{x}} dx + \int_1^{\infty} \frac{1}{e^{3x}} dx : \text{수렴} + \text{수렴} = \text{수렴} \end{aligned}$$

$$③ \int_0^1 \frac{\sin x}{x^{3/2}} dx \approx \int_0^1 \frac{x}{x^{3/2}} dx = \int_0^1 \frac{1}{x^{1/2}} dx : \text{수렴} (p < 1)$$

$$④ \int_0^1 \frac{\cos \pi x}{1-x} dx \approx \int_0^1 \frac{-1}{1-x} dx : \text{발산} (p=1)$$

$$\begin{aligned} 21. \gamma. \int_1^{\infty} (\sqrt{x^4+x+1} - \sqrt{x^4+1}) dx \\ &= \int_1^{\infty} \frac{(\sqrt{x^4+x+1} - \sqrt{x^4+1})(\sqrt{x^4+x+1} + \sqrt{x^4+1})}{(\sqrt{x^4+x+1} + \sqrt{x^4+1})} dx \end{aligned}$$

$$\begin{aligned} &= \int_1^{\infty} \frac{x}{\sqrt{x^4+x+1} + \sqrt{x^4+1}} dx \\ &\approx \int_1^{\infty} \frac{x}{2x^2} dx = \frac{1}{2} \int_1^{\infty} \frac{1}{x} dx : \text{발산} (p=1) \end{aligned}$$

$$ㄴ. \int_2^{\infty} \frac{\sin x}{x \log x} dx : \text{수렴} (\because \text{디리클레 판정법})$$

$$\begin{aligned} \text{ㄷ. } \int_0^1 x \sin\left(\frac{1}{x}\right) dx \left(\because \frac{1}{x} = t \right) \\ &= \int_1^{\infty} \frac{\sin t}{t^3} dt : \text{수렴} (\because \text{디리클레 판정법}) \end{aligned}$$

$$\begin{aligned} 22. ① \int_1^2 \frac{dx}{x^2-1} &= \int_1^2 \frac{1}{(x-1)(x+1)} dx \\ &\approx \frac{1}{2} \int_1^2 \frac{1}{(x-1)} dx : \text{발산} (p=1) \end{aligned}$$

$$② \int_2^{\infty} \frac{\ln x}{x^2} dx \approx \int_2^{\infty} \frac{1}{x^2} dx : \text{수렴} (p > 1)$$

$$③ \int_1^2 \frac{1}{\sqrt{x-1}} dx : \text{수렴} (p < 1)$$

$$④ \int_2^{\infty} \frac{dx}{x(\ln x)^2} : \text{수렴} (p > 1) \text{ (아래 공식 참고)}$$

$$23. \int_e^{\infty} \frac{1}{x(\ln x)^p} dx = \int_1^{\infty} \frac{1}{t^p} dt (\because \ln x = t) \text{ 이므로 } p > 1 \text{ 일 때 수렴한다.}$$

$$\begin{aligned} 24. \int_0^1 x^p \ln x dx &= \int_0^1 \frac{\ln x}{x^{-p}} dx \text{ 이므로 } -p < 1 \Leftrightarrow p > -1 \text{ 일 때} \\ &\text{수렴한다.} \end{aligned}$$

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11. 3번	12. 3번			

$$\begin{aligned}
 1. I &= \int_0^{\infty} \left(\frac{1}{\sqrt{x^2+9}} - \frac{a}{x+1} \right) dx \\
 &= \left[\sinh^{-1} \left(\frac{x}{3} \right) - a \ln(x+1) \right]_0^{\infty} \\
 &= \left[\ln \left(\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1} \right) - a \ln(x+1) \right]_0^{\infty} \\
 &= \left[\ln \left(\frac{\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1}}{(x+1)^a} \right) \right]_0^{\infty} \quad (a=1) \\
 &= \left[\ln \left(\frac{\frac{x}{3} + \sqrt{\frac{x^2}{9} + 1}}{(x+1)} \right) \right]_0^{\infty} = \ln \left(\frac{2}{3} \right) \text{이다.} \\
 \text{즉, } a=1, b=1, c=2, d=3 \text{이고 } a+b+c-d=1 \text{이다.}
 \end{aligned}$$

$$\begin{aligned}
 2. (가) \int_0^1 \frac{\cos x}{2x} dx &\approx \int_0^1 \frac{1}{2x} dx \text{ 이므로 발산한다.} \\
 (나) \int_{-\infty}^{-1} \frac{1}{\sqrt{3-x}} dx &= \int_4^{\infty} \frac{1}{\sqrt{t}} dt \quad (\because 3-x=t) \text{ 이므로 발산한다.} \\
 (다) \int_0^1 \frac{e^x}{\sqrt{2x}} dx &\approx \int_0^1 \frac{1}{\sqrt{2x}} dx \text{ 이므로 수렴한다.} \\
 (라) \int_0^1 \frac{\ln x}{1+x^3} dx &\approx \int_0^1 \ln x dx = -1 \text{ 이므로 수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 3. \neg. \int_0^1 \frac{x}{1-x} dx &\approx \int_0^1 \frac{1}{1-x} dx \text{ 이므로 발산한다.} \\
 \neg. \int_{-1}^1 \frac{1}{x^2} dx &\text{은 발산한다.} \\
 \neg. \int_{-2}^2 \frac{1}{\sqrt{4-x^2}} dx &= \left[\sin^{-1} \frac{x}{2} \right]_{-2}^2 = \pi \text{이므로 수렴한다.}
 \end{aligned}$$

[다른 풀이]

$$\begin{aligned}
 \int_{-2}^2 \frac{1}{\sqrt{4-x^2}} dx &= \int_{-2}^0 \frac{1}{\sqrt{(2-x)(2+x)}} dx + \int_0^2 \frac{1}{\sqrt{(2-x)(2+x)}} dx \\
 &\approx \int_{-2}^0 \frac{1}{\sqrt{4(2+x)}} dx + \int_0^2 \frac{1}{\sqrt{4(2-x)}} dx \\
 &= \text{수렴} + \text{수렴} = \text{수렴}
 \end{aligned}$$

$$\begin{aligned}
 \neg. \int_{-\infty}^{\infty} \frac{1}{(x^2+1)^2} dx &= \int_{-\infty}^0 \frac{1}{(x^2+1)^2} dx + \int_0^{\infty} \frac{1}{(x^2+1)^2} dx \text{ 이므로} \\
 \int_0^{\infty} \frac{1}{(x^2+1)^2} dx &\approx \int_0^{\infty} \frac{1}{x^4} dx \text{은 수렴한다.} \\
 \text{즉, 우함수이므로 } \int_{-\infty}^{\infty} \frac{1}{(x^2+1)^2} dx &= 2 \int_0^{\infty} \frac{1}{(x^2+1)^2} dx \text{은} \\
 &\text{수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 4. (\neg) \int_{-\infty}^{\infty} \frac{x}{1+x^4} dx &= \int_{-\infty}^0 \frac{x}{1+x^4} dx + \int_0^{\infty} \frac{x}{1+x^4} dx \text{ 이므로} \\
 \int_0^{\infty} \frac{x}{1+x^4} dx &\approx \int_0^{\infty} \frac{1}{x^3} dx \text{은 수렴한다.} \\
 \text{즉, 수렴하는 기함수이므로 } \int_{-\infty}^{\infty} \frac{x}{1+x^4} dx &= 0 \text{으로 수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 (\neg) \int_1^e \frac{1}{\ln x} dx &\text{은 } p=1 \text{이므로 발산한다.} \\
 (\neg) \int_{-1}^1 \frac{1}{x} dx &\text{은 발산한다.} \\
 (\neg) \int_3^{\infty} \frac{1}{x(\ln x)^2} dx &= \int_3^{\infty} \frac{1}{x(\ln x)^p} dx = \int_3^{\infty} \frac{1}{t^p} dt \quad (\because \ln x = t) \\
 \text{에서 } p=2 > 1 \text{이므로 수렴한다.}
 \end{aligned}$$

$$\begin{aligned}
 5. (\neg) \int_1^{\infty} \frac{\ln x}{x^3} dx &\approx \int_1^{\infty} \frac{1}{x^3} dx \text{은 수렴한다.} \\
 (\neg) \int_1^{\infty} \frac{\sin x}{x} dx &\text{은 디리클레 판정법에 의하여 수렴한다.} \\
 (\neg) \int_0^1 (\ln x)^2 dx &= (-1)^2 \times 2! \text{이므로 수렴한다.}
 \end{aligned}$$

[다른 풀이]

$$\int_0^1 (\ln x)^2 dx = \int_0^1 \frac{1}{(\ln x)^{-2}} dx \text{는 } p=-2 < 1 \text{이므로 수렴한다.}$$

[참고]

$$\int_0^1 (\ln x)^n dx = (-1)^n n!$$

$$\begin{aligned}
 6. (가) \int_{-\infty}^0 x^2 e^x dx &= - \int_{\infty}^0 t^2 e^{-t} dt \quad (\because x=-t) \\
 &= - \int_0^{\infty} \frac{t^2}{e^t} dt \approx - \int_0^{\infty} \frac{1}{e^t} dt \text{이므로 수렴한다.} \\
 (나) \int_1^{\infty} \frac{2+e^{-2x^2}}{1+\ln x} dx &\approx \int_1^{\infty} \frac{2}{\ln x} dx \quad (\text{특이점이 존재하지 않는다.}) \\
 \text{이때, } \ln x = t \text{로 놓으면 } \int_1^{\infty} \frac{2e^t}{t} dt &\text{이므로 발산한다.}
 \end{aligned}$$

$$\begin{aligned}
 (\neg) \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{\sin x}} dx &= \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{x - \frac{1}{3!}x^3 + \dots}} dx \\
 &= \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{1 - \frac{1}{3!}x^2 + \dots}} dx \\
 &\approx \int_0^{\frac{\pi}{3}} \frac{1}{x^2 \sqrt{x}} dx = \int_0^{\frac{\pi}{3}} \frac{1}{x^{\frac{5}{2}}} dx \text{이므로 발산한다.}
 \end{aligned}$$

$$(\neg) \int_0^{\frac{2}{3}} \frac{1}{\sqrt{|3x-1|}} dx = \int_0^{\frac{2}{3}} \frac{1}{|3x-1|^{\frac{1}{2}}} dx \text{은 수렴한다.}$$

$$\begin{aligned}
 7. (가) \int_1^{\infty} \frac{e^{-x^2}}{|x-3|^{1/2}} dx &= \int_1^4 \frac{e^{-x^2}}{|x-3|^{1/2}} dx + \int_4^{\infty} \frac{e^{-x^2}}{|x-3|^{1/2}} dx \\
 &\approx \int_1^4 \frac{e^{-9}}{|x-3|^{1/2}} dx + \int_4^{\infty} e^{-x^2} dx \text{ 일 때,} \\
 \int_1^4 \frac{e^{-9}}{|x-3|^{1/2}} dx &\text{은 수렴하고, } \int_4^{\infty} \frac{1}{e^{x^2}} dx \text{은 수렴하므로} \\
 \int_1^{\infty} \frac{e^{-x^2}}{|x-3|^{1/2}} dx &\text{은 수렴한다.} \\
 (나) \int_{-\infty}^{\infty} \frac{x}{(x^2+2)\ln(x^2+2)} dx &= \int_{-\infty}^0 \frac{x}{(x^2+2)\ln(x^2+2)} dx + \int_0^{\infty} \frac{x}{(x^2+2)\ln(x^2+2)} dx \text{ 일 때,} \\
 \int_0^{\infty} \frac{x}{(x^2+2)\ln(x^2+2)} dx &= \int_0^{\infty} \frac{x}{(x^2+2)\ln(x^2+2)} dx \quad (\because \text{특이점이 존재하지 않는다.})
 \end{aligned}$$

$$\approx \int_1^{\infty} \frac{x}{x^2 \ln(x^2)} dx = \int_1^{\infty} \frac{1}{2x \ln x} dx \text{은 발산한다.}$$

$$\text{따라서 } \int_{-\infty}^{\infty} \frac{x}{(x^2+2)\ln(x^2+2)} dx \text{은 발산한다.}$$

$$\text{다. } \int_{-\infty}^{\infty} x^4 e^{-x^2} dx = \int_{-\infty}^0 x^4 e^{-x^2} dx + \int_0^{\infty} x^4 e^{-x^2} dx \text{일 때,}$$

$$\int_0^{\infty} x^4 e^{-x^2} dx = \int_0^{\infty} \frac{x^4}{e^{x^2}} dx \approx \int_0^{\infty} \frac{1}{e^{x^2}} dx \text{은 수렴한다.}$$

$$\text{이때, 수렴하는 우함수이므로 } \int_{-\infty}^{\infty} x^4 e^{-x^2} dx = 2 \int_0^{\infty} x^4 e^{-x^2} dx \text{은 수렴한다.}$$

$$\text{라. } \int_{-\infty}^{\infty} x^{100} e^{-x^2} dx = \int_{-\infty}^0 x^{100} e^{-x^2} dx + \int_0^{\infty} x^{100} e^{-x^2} dx \text{일 때,}$$

$$\int_0^{\infty} x^{100} e^{-x^2} dx = \int_0^{\infty} \frac{x^{100}}{e^{x^2}} dx \approx \int_0^{\infty} \frac{1}{e^{x^2}} dx \text{이므로 수렴하고,}$$

$$\int_{-\infty}^0 x^{100} e^{-x^2} dx = - \int_0^{\infty} t^{100} e^{-t^2} dt = \int_0^{\infty} \frac{t^{100}}{e^{t^2}} dt \approx \int_0^{\infty} \frac{1}{e^{t^2}} dt$$

이므로 수렴한다.

$$\text{즉, } \int_{-\infty}^{\infty} x^{100} e^{-x^2} dx \text{은 수렴+수렴이므로 수렴한다.}$$

$$8. \text{ 가. } \int_0^1 \frac{x}{\ln x} dx \approx \int_0^1 \frac{1}{\ln x} dx \text{은 발산한다.}$$

$$\text{나. } \int_0^1 \frac{1}{\ln(2x)} dx = \frac{1}{2} \int_0^2 \frac{1}{\ln t} dt \quad (\because 2x=t) \text{은 발산한다.}$$

$$\text{다. } \int_0^1 \frac{1}{x \ln x} dx = \int_0^1 \frac{1}{e^{-t} t} dt \quad (\because \ln x = -t) \\ = - \int_0^1 \frac{1}{t} dt = - \left(\int_0^1 \frac{1}{t} dt + \int_1^{\infty} \frac{1}{t} dt \right) \text{은 발산한다.}$$

$$\text{라. } \int_0^1 \frac{1}{x(\ln(2x))^2} dx = \int_0^2 \frac{1}{t(\ln t)^2} dt \quad (\because 2x=t)$$

$$= \int_{-\infty}^{-\ln 2} \frac{1}{e^{-\theta}(-\theta)^2} e^{-\theta} d\theta \quad (\because \ln t = -\theta)$$

$$= \int_{-\ln 2}^{\infty} \frac{1}{\theta^2} d\theta = \int_{-\ln 2}^1 \frac{1}{\theta^2} d\theta + \int_1^{\infty} \frac{1}{\theta^2} d\theta$$

은 발산+수렴이므로 발산한다.

$$9. \text{ 가. } \int_1^{\infty} \frac{\sin x}{x} dx \text{은 디리클레 판정법에 의하여 수렴한다.}$$

$$\text{나. } \int_0^1 \frac{1}{\sqrt{x}} \cos \frac{1}{x} dx = \int_{\infty}^1 \sqrt{t} \times \cos t \times -\frac{1}{t^2} dt \quad (\because \frac{1}{x}=t) \\ = \int_1^{\infty} \frac{\cos t}{t^{\frac{3}{2}}} dt \text{이므로 디리클레 판정법에 의하여 수렴한다.}$$

$$\text{다. } \int_{\pi}^{\infty} \sin(x^2) dx \text{은 } p > 1 \text{이므로 수렴한다.}$$

$$\text{라. } \int_0^{\infty} \frac{\sqrt{x}}{x+x^2} dx = \int_0^1 \frac{\sqrt{x}}{x+x^2} dx + \int_1^{\infty} \frac{\sqrt{x}}{x+x^2} dx \text{일 때,}$$

$$\int_0^1 \frac{\sqrt{x}}{x+x^2} dx = \int_0^1 \frac{\sqrt{x}}{x(1+x)} dx \approx \int_0^1 \frac{\sqrt{x}}{x} dx = \int_0^1 \frac{1}{\sqrt{x}} dx$$

이므로 수렴하고,

$$\int_1^{\infty} \frac{\sqrt{x}}{x+x^2} dx \approx \int_1^{\infty} \frac{\sqrt{x}}{x^2} dx = \int_1^{\infty} \frac{1}{x^{\frac{3}{2}}} dx \text{이므로 수렴한다.}$$

$$\text{즉, } \int_0^{\infty} \frac{\sqrt{x}}{x+x^2} dx \text{은 수렴+수렴이므로 수렴한다.}$$

$$10. \text{ a. } \int_1^{\infty} \frac{|\sin x|}{x} dx \approx \int_1^{\infty} \frac{1}{x} dx \text{은 발산한다.}$$

$$\text{b. } \int_1^9 \frac{1}{\sqrt[3]{x-1}} dx = \int_1^9 \frac{1}{(x-1)^{\frac{1}{3}}} dx \text{은 수렴한다.}$$

$$\text{c. } \int_0^{\infty} \frac{\arctan(e^x)}{2+e^x} dx \approx \int_0^{\infty} \frac{1}{e^x} dx \text{은 수렴한다.}$$

$$\text{d. } \int_0^1 \frac{\ln x}{x} dx \text{은 발산한다.}$$

$$11. \text{ a. } \int_1^2 \frac{x+1}{x\sqrt{x-1}} dx \approx \int_1^2 \frac{2}{(x-1)^{\frac{1}{2}}} dx \text{은 수렴한다.}$$

$$\text{b. } \int_0^1 \frac{1}{\sqrt{x^2+2x}} dx = \int_0^1 \frac{1}{\sqrt{x(x+2)}} dx \approx \int_0^1 \frac{1}{\sqrt{2x}} dx \text{은 수렴한다.}$$

$$\text{c. } \int_1^{\infty} \frac{1}{x \ln x} dx \text{은 발산한다.}$$

$$\text{d. } \int_{-\infty}^{\infty} \operatorname{sech} x dx = \int_{-\infty}^{\infty} \frac{2}{e^x + e^{-x}} dx$$

$$= \int_{-\infty}^0 \frac{2}{e^x + e^{-x}} dx + \int_0^{\infty} \frac{2}{e^x + e^{-x}} dx \text{일 때,}$$

$$\int_0^{\infty} \frac{2}{e^x + e^{-x}} dx \approx \int_0^{\infty} \frac{2}{e^x} dx \text{이므로 수렴한다.}$$

따라서 수렴하는 우함수이므로

$$\int_{-\infty}^{\infty} \frac{2}{e^x + e^{-x}} dx = 2 \int_0^{\infty} \frac{2}{e^x + e^{-x}} dx \text{은 수렴한다.}$$

$$12. \text{ a. } \int_2^{\infty} e^{-\sqrt{\ln x}} dx = \int_{\ln 2}^{\infty} e^{-\sqrt{t}} e^t dt \quad (\because \ln x = t)$$

$$= \int_{\ln 2}^{\infty} e^{t-\sqrt{t}} dt \approx \int_{\ln 2}^{\infty} e^t dt \text{은 발산한다.}$$

$$\text{b. } \int_{-1}^0 \frac{e^x}{x^3} dx = - \int_1^{\infty} t e^{-t} dt \quad (\because \frac{1}{x} = -t)$$

$$= - \int_1^{\infty} \frac{t}{e^t} dt \approx - \int_1^{\infty} \frac{1}{e^t} dt \text{은 수렴한다.}$$

$$\text{c. } \int_0^{\pi} \frac{x}{1-\cos x} dx = \int_0^{\pi} \frac{x}{1-\left(1-\frac{1}{2!}x^2+\frac{1}{4!}x^4-\dots\right)} dx$$

$$= \int_0^{\pi} \frac{x}{\frac{1}{2!}x^2-\frac{1}{4!}x^4+\dots} dx = \int_0^{\pi} \frac{x}{x^2\left(\frac{1}{2!}-\frac{1}{4!}x^2+\dots\right)} dx$$

$$\approx \frac{1}{\frac{1}{2!}} \int_0^{\pi} \frac{x}{x^2} dx = 2! \int_0^{\pi} \frac{1}{x} dx \text{은 발산한다.}$$

[다른 풀이]

$$\int_0^{\pi} \frac{x}{1-\cos x} dx = \int_0^{\pi} \frac{x^2}{1-\cos x} \cdot \frac{1}{x} dx$$

$$\approx 2 \int_0^{\pi} \frac{1}{x} dx \text{는 발산한다. } \left(\because \lim_{x \rightarrow 0} \frac{1-\cos x}{x^2} = \frac{1}{2} \right)$$

$$\text{d. } \int_0^{\frac{\pi}{2}} \frac{x}{\sqrt{\cos^3 x}} dx = - \int_{\frac{\pi}{2}}^0 \frac{\frac{\pi}{2}-t}{\sqrt{\sin^3 t}} dt \quad (\because \frac{\pi}{2}-x=t)$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}-t}{\sqrt{\sin^3 t}} dt$$

$$\approx \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{\sin^3 t}} dt = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{\left(t-\frac{1}{3!}t^3+\dots\right)^3}} dt$$

$$= \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{t^3\left(1-\frac{1}{3!}t^2+\dots\right)^3}} dt \approx \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{\sqrt{t^3}} dt = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}}{t^{\frac{3}{2}}} dt \text{이므로}$$

발산한다.

1. $p < 1, -\frac{1}{(1-p)^2}$	2. 1번	3. 4번	4. 2번
5. 3번	6. 3번	7. 2번	8. 2번
9. 3번			

$$1. I = \int_1^e \frac{\ln(\ln x)}{x(\ln x)^p} dx = \int_0^1 \frac{\ln t}{t^p} dt \quad (\because \ln x = t)$$

이므로 수렴하기 위한 $p < 1$ 이다.

$$\therefore I = \int_0^1 \frac{\ln t}{t^p} dt = \left[\frac{1}{1-p} t^{1-p} \ln t - \frac{1}{(1-p)^2} t^{1-p} \right]_0^1$$

$$= -\frac{1}{(1-p)^2}$$

$$2. \int_0^\infty \left(\frac{1}{\sqrt{x^2+4}} - \frac{k}{x+2} \right) dx$$

$$= \left[\sinh^{-1} \frac{x}{2} - k \ln(x+2) \right]_0^\infty$$

$$= \left[\ln \left(\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1} \right) - \ln(x+2)^k \right]_0^\infty$$

$$= \left[\ln \frac{\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1}}{(x+2)^k} \right]_0^\infty \quad (k=1)$$

$$= \left[\ln \frac{\frac{x}{2} + \sqrt{\frac{x^2}{4} + 1}}{(x+2)} \right]_0^\infty = -\ln \frac{1}{2} = \ln 2$$

$$3. \int_0^\infty \frac{e^{\beta x^2}}{x^\alpha} dx = \int_0^1 \frac{e^{\beta x^2}}{x^\alpha} dx + \int_1^\infty \frac{e^{\beta x^2}}{x^\alpha} dx \quad \text{일 때,}$$

$$\int_0^1 \frac{e^{\beta x^2}}{x^\alpha} dx \approx \int_0^1 \frac{1}{x^\alpha} dx \quad \text{은 } \alpha < 1 \text{ 이면 수렴하며,}$$

$$\int_1^\infty \frac{e^{\beta x^2}}{x^\alpha} dx \int_1^\infty e^{\beta x^2} dx \quad \text{은 } \beta < 0 \text{ 이면 수렴한다.}$$

따라서 $\alpha < 1, \beta < 0$ 이다. 즉, (가), (나)는 참이다.

(다) [반례]

$$\alpha = 0.5, \beta = -0.1 \text{ 이면 } \int_0^\infty \frac{e^{\beta x^2}}{x^\alpha} dx \text{ 는 수렴하지만}$$

$\alpha + \beta > 0$ 이다. 따라서 (다)는 거짓이다.

$$4. \int_1^2 \frac{x^x - x}{(x-1)^p} dx \quad (x-1=t \text{ 로 치환})$$

$$= \int_0^1 \frac{(t+1)^{t+1} - (t+1)}{t^p} dt$$

$$= \int_0^1 \frac{(t+1)^t \{(t+1)^t - 1\}}{t^p} dt$$

$$= \int_0^1 \frac{(t+1)^t - 1}{t^p} dt$$

$$\text{이때, } (t+1)^t = e^{t \ln(t+1)}$$

$$= e^{t \left(t - \frac{1}{2}t^2 + \frac{1}{3}t^3 + \dots \right)}$$

$$= e^{t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots}$$

$$= 1 + \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right) + \frac{1}{2!} \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right)^2 + \dots$$

이다.

$$\int_0^1 \frac{(t+1)^t - 1}{t^p} dt$$

$$= \int_0^1 \frac{1}{t^p} \left(1 + \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right) + \frac{1}{2!} \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right)^2 + \dots - 1 \right) dt$$

$$= \int_0^1 \frac{1}{t^p} \left(\left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right) + \frac{1}{2!} \left(t^2 - \frac{1}{2}t^3 + \frac{1}{3}t^4 + \dots \right)^2 + \dots \right) dt$$

$$= \int_0^1 \frac{t^2}{t^p} \left(1 - \frac{1}{2}t + \dots \right) dt$$

$$\approx \int_0^1 \frac{t^2}{t^p} dt$$

$$= \int_0^1 \frac{1}{t^{p-2}} dt$$

따라서 $p-2 < 1 \Leftrightarrow p < 3$ 일 때 수렴하므로 가장 큰 자연수는 2이다.

$$5. \Gamma\left(\frac{1}{2}\right) = \int_0^\infty t^{-\frac{1}{2}} e^{-t} dt = \int_0^\infty \frac{1}{\sqrt{t}} e^{-t} dt$$

$$= \int_0^\infty \frac{1}{u} e^{-u^2} 2u du \quad (\because \sqrt{t} = u)$$

$$= 2 \int_0^\infty e^{-u^2} du$$

$$= \sqrt{\pi} \quad \left(\because \int_0^\infty e^{-u^2} du = \frac{\sqrt{\pi}}{2} \right)$$

$$6. \textcircled{1} \Gamma(n+1) = n\Gamma(n) : \text{참}$$

$$\textcircled{2} n \text{ 은 자연수이면 } \Gamma(n+1) = n! : \text{참}$$

$$\textcircled{3} \Gamma(x) = \int_0^\infty t^{x-1} e^{-t} dt = 2 \int_0^\infty s^{2x-1} e^{-s^2} ds \quad (\because t = s^2) : \text{거짓}$$

$$\textcircled{4} \frac{\Gamma(5/2)}{\Gamma(1/2)} = \frac{(3/2)\Gamma(3/2)}{\Gamma(1/2)} = \frac{(3/2)(1/2)\Gamma(1/2)}{\Gamma(1/2)} = \frac{3}{4} : \text{참}$$

$$7. \textcircled{1} f(1) = \int_0^\infty e^{-x} dx = \Gamma(1) = 0! = 1 : \text{참}$$

$$\textcircled{2} f(2) = \int_0^\infty x e^{-x} dx = \Gamma(2) = 1! = 1 : \text{거짓}$$

$$\textcircled{3} \lim_{\alpha \rightarrow 0+} f(\alpha) = \infty : \text{참} (\because \text{감마함수 그래프 참고})$$

$$\textcircled{4} \lim_{\alpha \rightarrow \infty} f(\alpha) = \infty : \text{참} (\because \text{감마함수 그래프 참고})$$

$$8. A(n) = (n-1)A(n-1) \text{ 이므로}$$

$$\frac{A(n)}{A(n-1)} + \frac{A(n-1)}{A(n-2)} + \frac{A(n-2)}{A(n-3)} + \dots + \frac{A(2)}{A(1)}$$

$$= \frac{(n-1)A(n-1)}{A(n-1)} + \frac{(n-2)A(n-2)}{A(n-2)} + \dots + \frac{1A(1)}{A(1)}$$

$$= (n-1) + (n-2) + (n-3) + \dots + 1$$

$$= \frac{n(n-1)}{2}$$

$$9. \int_0^1 (-\ln t)^5 dt = - \int_0^1 (\ln t)^5 dt$$

$$= -(-1)^5 \times 5! = 120$$

[참고] 자연수 n 에 대하여 $\int_0^1 (\ln x)^n dx = (-1)^n n!$ 이 성립한다.